

REAL TIME FACE RECOGNITION AND TRACKING SYSTEM

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ABSTRACT

Nowadays that security comes into more prominence every day, it is necessary for people to keep more passwords in their mind and carry more cards with themselves. Such implementations however, are becoming less secure and practical, thus leading to an increasing interest in techniques related to biometrics systems. Biometrics systems are the systems which store physical properties of people in electronic environment and enable them to be recognized by the stored electronic information when needed. Biometrics is the identification of human. It works on the principle of identification of physical properties of a person which he or she cannot alter, are distinctive from others, can be used for identification, and are in his or her possession only. Extensive studies are conducted on biometrics techniques such as fingerprint, hand, face, iris, retina and voice recognition. Some systems have been developed, tested, and results have been obtained. Face recognition systems are among the most important subjects in biometrics systems. These systems, which are very important for security in particular, have been widely used and developed in many countries. This study aims to achieve face recognition successfully by detecting human face in real time, based on Principal Component Analysis (PCA) algorithm and comparing the result with pre-recorded face samples.

Index Terms— Principal component analysis, Face recognition, Information security system, PCA, Face tracking

1. INTRODUCTION

Today, biometric systems are used in many fields. Depending on their field of practice, these systems may include some problems. With the aim of providing security in biometric systems, different features can be used. These features are the characteristics such as fingerprints, iris, retina, palm and face, which differ from people to people. Systems which are using any of these are called biometric systems, structures which are using two or more of these are called multiple biometric systems [1].

Biometric face recognition system is used for security in banks, airport check-in, border crossings, prisons, security practice, (criminal identification – age equation), homes, offices, factories, sport complexes and so forth. Biometric face recognition system works like that: the face is scanned by the camera, significant points on the face are obtained as data, and then the data collected is compared to the records on the system databases [2].

In this study, a real time biometric system which is based on face recognition was studied. The system as an example is applied to personnel automation.

2. FACE RECOGNITION SYSTEMS

Face recognition systems' senses of work are exactly the same with the biometric recognition systems. Face recognition system is based on the idea that each human being is different and unique in creation. If we elucidate this, facial structure has parts that are unique to each person like fingerprints. But people do not know about this unique facial structure. Each person has his own facial structure, these lines known as the symmetry of the faces are matched in computer environment and face recognition process takes place. As a result of the biometric matching carried out in computer environment, facial structure is compared to the faces on the database and the personnel who has the matching facial structure is allowed to log in.

In card reading systems, people may give their cards to other people and may lose their cards. But in face recognition systems, it is out of the question. Fingerprints reading systems may cause problems in reading the fingerprints in some sectors. Metal industry and firms where chemical materials are used, chemical material waste and skin irritation on hands make it hard for the system to read and therefore prevent such a system from working properly. With face recognition system, all the negativity is done away with, and safer and more useful access control products are put into service.

Face recognition systems that can detect a large number of faces can have many practices. For instance, ATM and credit card systems are interested in these systems in order to stop misuse and fraud. One way of achieving this is to put the picture of the owner on the credit card. Face recognition system can recognize the picture of the credit card owner

and offer high level security. Security and enquiry institutions are preoccupied with the development of rapid and efficient methods of recognizing people. For some applications, facial vision is the only source of data that is appropriate for recognizing the face. One example of this is the picture taken from the video image. In such an image, facial vision may not even be seen partially. Face recognition which is combined with voice recognition can be used in the interfaces which are between human and computer. In this case, voice information can be used to limit searching and therefore face recognition algorithm is applied to a reduced amount of images. Such hybrid methods yield more reliable results [4].

3. PRINCIPAL COMPONENT ANALYSIS

PCA (Principal Component Analysis) is based on the comparison and compression of the image in a way that detects the similar parts and leaves specific parts of it. [1] In PCA method, the examples taken should be same size with the ones on the database for the recognition process to take place. Pictures that are used in this method are present as minimized and compressed on the database. Thus, database load is decreased and face recognition speed is accelerated. [2].

Principal component analysis is a method of analysis that can gain significant data from large scale data set. PCA can present the similarities and differences in a large scale data set. In large scale data set, analysis process is difficult and the margin of error is high. In this respect, PCA have considerable importance for many systems [3].

PCA is a statistical method that is often used in computer vision and pattern recognition systems. Sirovich and Kirby were first to use PCA in representing human faces. They proved that facial images can be obtained approximately [3]. PCA is about finding the vector that gives the distribution of facial images in the image space in the best way. These vectors are the eigenvectors of covariance matrix corresponding to original facial images. Vectors are expressed as Eigen face. The reason why it is expressed as Eigen face is as an image it resembles to facial image [3].

PCA'S practice is like that: the transition of facial images on the database into vectors such as $\Gamma_1, \Gamma_2, \dots, \Gamma_M$ is made. Here an $N \times N$ size facial image corresponds to a $N^2 \times 1$ size vector. With these vectors, $M \times N^2$ size X data set is acquired. In the acquired X data set, the average vector is calculated. Calculated average vector is subtracted from each vector on matrix $M \times N^2$ (2).

$$y = \frac{1}{M} \sum_{i=1}^M \Gamma_i, \quad i = 1, \dots, M \quad (1)$$

$$f_i = \Gamma_i - y \quad (2)$$

$$A = [f_1, f_2, \dots, f_M] \quad (3)$$

After the subtraction of average matrix from all the vectors in the data set matrix A (3) is acquired. Matrix A contains $N^2 \times 1$ size M piece vector. Then, for each f_i data set on matrix A, $N^2 \times 1$ size h_i vector, which represents this data set best, is acquired. Each h_i vector should be given in a way that will maximize g_i coefficients that are defined in formula (4). h_i vectors and g_i coefficients are the eigenvectors and Eigen coefficients of the covariance matrix C that is given in formula (5).

$$g_i = \frac{1}{M} \sum_{i=1}^M (h_i^T f_i)^2 \quad (4)$$

$$C = AA^T \quad C_{i,j} = f_i f_j^T \quad (5)$$

Each facial image in education set has $N \times N$ size. This means covariance matrix C has $N^2 \times N^2$ size. As the size of the picture increases, the size of covariance matrix C increases squarely. This makes it hard for Eigen values and Eigen vectors to be calculated. For this reason, the size of covariance matrix C needs to be reduced. The numbers of images in the education set are smaller than the size of the pictures. ($M < N^2$) This demonstrates that there are M piece significant Eigen vector instead of N^2 Eigen vector. In this case, Eigen values that correspond to Eigen vectors will be zero. In this way, the size of the covariance matrix, whose Eigen values and Eigen vector will be calculated, is reduced [3].

As shown in formula (6) $M \times M$ size covariance matrix L is calculated and as shown in formula (7) h_i Eigen vectors each of whom $M \times 1$ size and g_i Eigen value which corresponds to each Eigen vector are calculated [3].

$$L = A^T A \quad L_{i,j} = f_i^T f_j \quad (6)$$

$$A^T A_{vi} = g_i v_i \quad (7)$$

$$AA^T A_{vi} = g_i A_{vi} \quad (8)$$

As seen in the equations (7, 8) v_i 's are the Eigen vectors of $A^T A$, $A v_i$ are the Eigen vectors of AA^T . In this way, M real Eigen vector is acquired, each of whom is $N^2 \times 1$ size, by forming the combination of the facial images in the education set and v_i .

$$u_i = \sum_{k=1}^M v_{i,k} f_k \quad i = 1, \dots, M \quad (9)$$

Out of M piece Eigen vectors, whose acquisition is shown in formula (9), M' piece is important. Other Eigen vectors are disregarded (10).

$$\sum_{i=1}^{M'} g_i / \sum_{i=1}^M g_i > 0.90 \quad (10)$$

Acquired Eigen vectors are used in producing feature vectors of the images in the education set. As seen in formula (11) and formula (12) feature vectors that belong to the image i . are calculated. Calculated feature vectors consist of M piece number.

$$w_k = h_k^T f_i \quad (11)$$

$$l_i = [w_1, w_2, \dots, w_M] \quad i = 1, \dots, M \quad (12)$$

After the subtraction of feature vectors of all facial images in the education set, these feature vectors are recorded on the database. In order to find out which image in the education set is similar to a Γ image taken as entry for the test process, feature vectors of the test image are calculated as shown in formula (13) and (14).

$$w_k = h_k^T (\Gamma - y) \quad k = 1, \dots, M \quad (13)$$

$$l = [w_1, w_2, \dots, w_M] \quad (14)$$

Feature vector that is acquired for the test image is compared to the feature vector that is produced for the education images. Comparison process is carried out whether the distance between two vectors is small or big with respect to a determined ϵ_k threshold value.

$$\frac{\|l - l_k\|}{l_k} \leq \epsilon_k \quad k = 1, \dots, M \quad (15)$$

$$\Gamma' = y + \sum_{i=1}^M w_i h_i \quad (16)$$

$$e = \frac{\|\Gamma' - \Gamma\|}{\|\Gamma\|} \quad (17)$$

Eigen faces are used in order to regain any image in the education set. Approximate production of any facial image in the education set is shown in formula (16). Error rate that can occur in the process of regaining the image is as it is in formula (17) [3].

4. MATERIAL AND APPLICATION

Real time face recognition and tracking system practice is basically a system that uses PCA algorithm. System C# is developed with the use of programming language and the library of OpenCvsharp and EmguCV. System can be used with different purposes. In this study, it is applied as a biometric system in a real time personnel tracking system for ensuring the safe entry of personnel.

System is developed by taking Model-View-Controller (MVC) architectural pattern. MVC is based on the abstraction of data and display in complicated applications, where a large amount of data is presented to user. Thus, data (model) and the user interface (view) can be organized

without affecting each other. MVC solves this with the component called *controller* by means of subtracting data access and business logic from data display and user interaction.

In the system, interface is on the front layer, face detection and recognition systems are on the tier, recording processes are on the boundary layer.

DevExpress is used in designing the interface of the software. DevExpress is a component that is developed for .Net platform. When the software is run, necessary DevExpress libraries should be installed in the computers where the system will be installed so that relevant interface components don't give error. Every time the system is run, it first checks whether it is the recorded face. Besides, when the system launches, it searches the camera device, if any device is not found, it gives a warning. System is launched by the administrator. System's first function is the detection and recognition of the human face. This function is carried out by using the methods and classes in the EmguCV library. EmguCV uses *HaarCascade* class for detecting the human face and eye. This class takes values and information necessary for the detection of human face from an XML file offered for free distribution.

Real time face recognition and tracking system in this study can detect more than one face in its field of view and track it. The system presents the following features in general.

- Detection of human face
- Detection of eyes
- Framing the detected face
- Tracking the detected face and eye (the frame moves with the face)
- Checking the detected face that it is not recorded before.
- Preventing the person from logging in when the matching record is not found.
- Captioning the circled face when the matching record is found.
- Letting the person recognized by the system log in.
- Recording the date and time information of the person who logs in/logs out at that time.

In Fig. 1, the flowchart of the system is presented. Firstly, the system is launched and then accessed so as to use the suitable camera in the system by checking the camera devices connected to the computer. System will start after it accesses to camera without problem. When the system starts, it will give a warning 'there is no record in system' after instalment since it is checking the records. A short time after system is launched; camera detects human face and eye in front of it. It will put the detected face in a big circle, the detected eyes in small circle; it will show a proof of face detection. If the face moves, it follows its movement in order to process on the circled face. Afterwards, it scans whether

such a face is recorded by searching the detected face in the previous records. While doing this process, it uses PCA algorithm. When the matching record is not found, persons can be prevented from logging into system. When the matching record is found, user information recorded in the system is written on the face circled beforehand. Personal recognized by the system is allowed to logging into system. Finally, time and day information of the personnel logging in at that time are recorded.

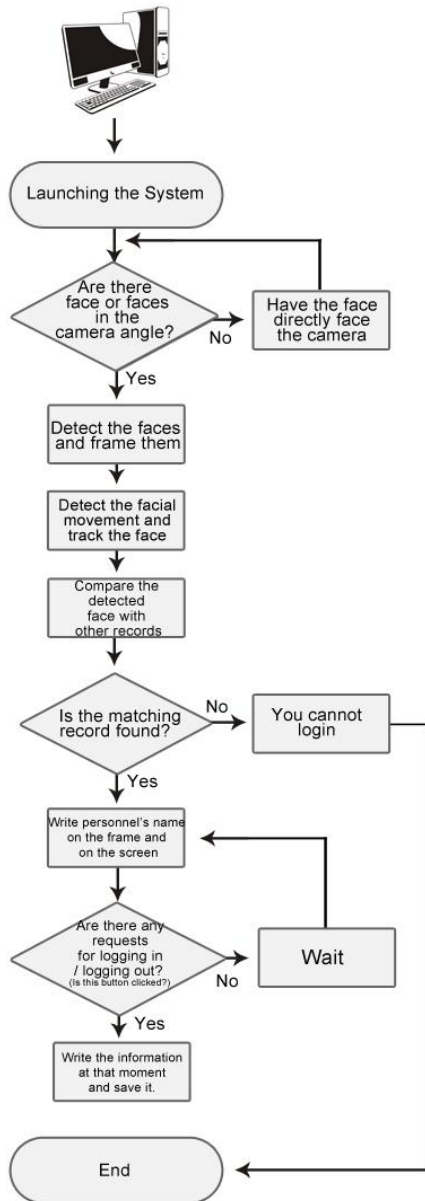


Fig. 1 Flowchart diagram of the system

In Fig. 2, a block diagram, which shows the way the system runs, is presented. In Fig. 3, system use-case diagram is presented. All the actors in the system and the processes they can carry out upon the system are seen. In Fig. 4, some of the faces on system are presented. In Fig. 5, the

screenshot of the moment during the face detection and tracking process in the program interface. As can be seen in the figure, when the faces and eyes are detected, they are put into squares and these squares move with the face and the eyes, that is the system automatically carries out real time tracking. In Fig. 6, face recognition process is demonstrated. Faces in the system database and detected faces can be matched and when the recognition process takes place, the system detects the relevant person and shows his/her personal information.

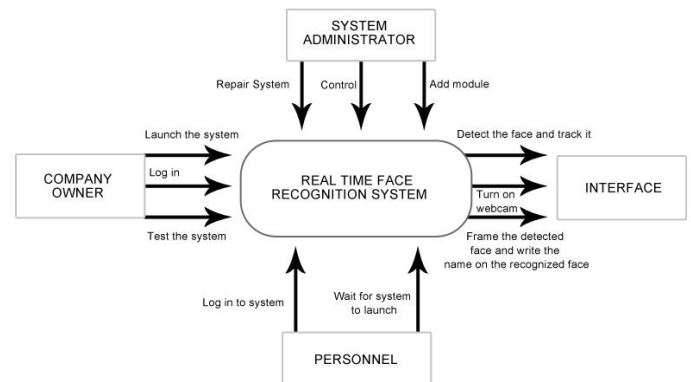


Fig. 2 System block diagram

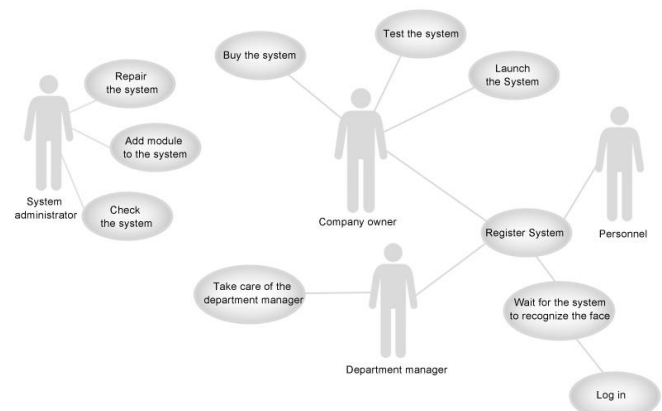


Fig. 3 System use-case diagram

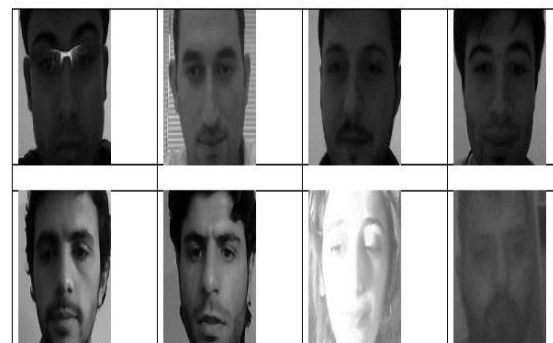


Fig. 4 Profiles from face database

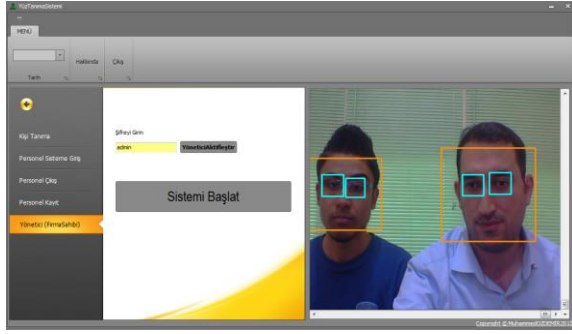


Fig. 5 Face detection and tracking

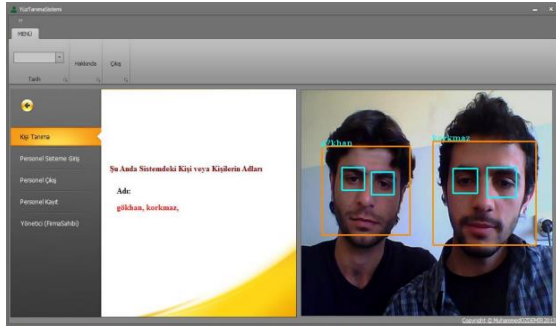


Fig. 6 Face recognition and tracking

5. CONCLUSION

Biometric systems are structures that have been often used in recent years. This biometric system is based on the use of some physiological features of the person for security. It is predicted that biometric systems will be an indispensable part of the information security systems in the coming years. Thus, people won't have to carry cards and/or memorize their own passwords.

In this study, a biometric system for human face detection and recognition is put into practice. This system put into practice works real time and successfully carries out face recognition, detection and tracking. In this way, the system has the feature of being integrated into different practices. In this study, face detection and tracking system, is put into practice in a personal automation. Here, the aim is to carry out personnel's entry and exit automatically and safely. In this system practiced basing on PCA, in the future studies, various optimization algorithms can be used in order to increase system performance.

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